IGCSE 0607 Practice Questions — E1.3 — Combined Questions & Markscheme

Powers and roots

Total Questions: 6 — **Total Marks:** 34 Marks

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- **E1.3** Powers and roots
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(2 Qs)*

1 **(a)** Work out the value of

$$\sqrt{169} + 2^3$$

..... **[2]**

(b) Work out the value of

$$4^2 \times \sqrt[3]{125}$$

..... **[2]**

2 Solve the following equations.

(a)

$\sqrt[3]{3x+4} = 4$

$x =$ ..... **[3]**

(b)

$x^2 = 4^3 + 6^2$

$x =$ ..... , or $x =$ ..... **[2]**

(c)

$\sqrt{x} + 2\sqrt{x} = 21$

$x =$ ..... **[2]**

3 **(a)** Calculate the value of

$$\frac{\sqrt[3]{450} + 12.5^2}{3.4^3 - \sqrt{80}}$$

Write down all the figures on your calculator display.

..... **[2]**

(b) Write your answer to part (a) correct to 3 decimal places.

..... **[1]**

(c) Find the largest integer n such that $n^4 < 2000$.

$n =$ ..... **[2]**

4 **(a)** Write down the value of

$\sqrt{196}$

..... **[1]**

(b) Work out the value of

$4^3 - \sqrt[3]{125}$

..... **[2]**

(c) Calculate the exact value of

$\sqrt{13^2 - 12^2}$

..... **[2]**

5 **(a)** **(i)** Calculate the value of

$$\sqrt[3]{8.5^2 - 4.1^3}$$

Write down your full calculator display.

..... **[2]**

(ii) Write your answer to part (a)(i) correct to 3 significant figures.

..... **[1]**

(b) Calculate the exact value of

$$\frac{4.8^2}{\sqrt{0.0576}}$$

..... **[2]**

6 **(a)** Find the integer k such that

$$\sqrt[3]{k} = 4^2 - 3^2 - 2$$

$k =$ ..... **[2]**

(b) Calculate the exact value of

$$\frac{\sqrt{10^2 - 6^2}}{\sqrt[3]{125} - 3^2}$$

..... **[3]**

(c) Given that $y > 0$ and $\sqrt[3]{y} + \sqrt[3]{y} = 6$, work out the value of y^2.

..... **[3]**

1 **(a)** 21 **[B2]**

M1 for 13 or 8

(b) 80 **[B2]**

M1 for 16 or 5

[Back to Question] *[Total Marks: 4]*

2 **(a)** $3x + 4 = 64$ **[M1]**

$3x = 60$ **[M1]**

20 **[A1]**

(b) $x^2 = 64 + 36 = 100$ **[M1]**

$10, -10$ **[A1]**

(c) $3\sqrt{x} = 21$ or $\sqrt{x} = 7$ **[M1]**

49 **[A1]**

[Back to Question] *[Total Marks: 7]*

3 **(a)** 5.3990299... **[B2]**

M1 for numerator evaluated to 163.91... or denominator evaluated to 30.359...

(b) 5.399 **[B1]**

(c) 6 **[B2]**

M1 for evaluating $6^4 = 1296$ and $7^4 = 2401$

[Back to Question] *[Total Marks: 5]*

4 **(a)** 14 **[B1]**

(b) 59 **[B2]**

If 0 scored, M1 for 64 or 5 seen **[M1]**

(c) 5 **[B2]**

If 0 scored, M1 for $169 - 144$ or 25 seen **[M1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** **(i)** 1.49315... **[B2]**

If 0 scored, M1 for 3.329 seen or 72.25 and 68.921 **[M1]**

(ii) 1.49 **[B1]**

(b) 96 **[B2]**

If 0 scored, M1 for 23.04 or 0.24 seen **[M1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** 125 **[B2]**

If 0 scored, M1 for evaluating $16 - 9 - 2$ to get 5 **[M1]**

(b) -2 **[B3]**

M1 for $\sqrt{64}$ or 8 **[M1]**

M1 for $5 - 9$ or -4 **[M1]**

(c) 729 **[B3]**

M1 for $2\sqrt[3]{y} = 6$ or $\sqrt[3]{y} = 3$ **[M1]**

M1 for $y = 27$ **[M1]**

[Back to Question] *[Total Marks: 8]*